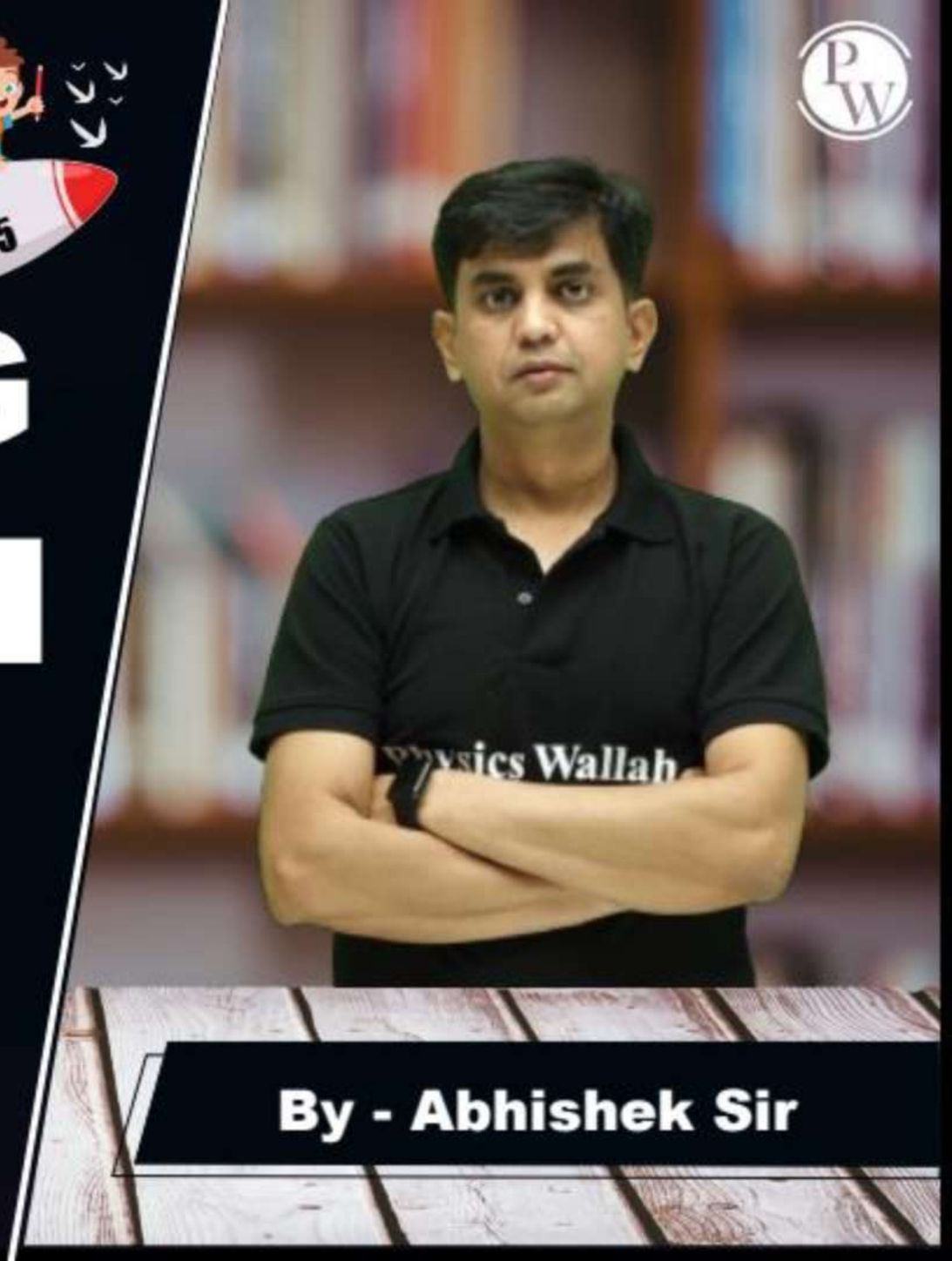
CS & IT ENGINEERING

Computer Network

Transport Layer

Lecture No. - 06



Recap of Previous Lecture

Topic TCP Connection Establishment

Topic TCP Connection Release

Topics to be Covered

- Topic: Socket Programming
- Topic: TCP Socket
- Topic: UDP Socket

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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Topic : TCP Segment Structure

0	16	31
Source Port No. (16 bit)	Dest. Port No. (16 bit)	
Sequence Number (32 bit)		
Acknowledgement Number (32 bit)		
HLEN	URG ACK PSH RST SYN FIN	Window Size (16 bit)
Checksum (16 bit)		Urgent Pointer (16 bit)
Options [Optional Header]		
—//— Payload —//—		

TCP Header (5 to 15 word) (20 to 60 byte)

TCP Base Header 5 word (20 byte)

0 to 10 word (0 to 40 byte)

HLEN →

#Q. Consider a TCP client and a TCP server running on two different machines. After completing data transfer, the TCP client calls close to terminate the connection and a FIN segment is sent to the TCP server. Server-side TCP responds by sending an ACK which is received by the client-side TCP. As per the TCP connection state diagram (RFC 793), in which state does the client side TCP connection wait for the FIN from the server-side TCP?

[GATE-2017, Set-1, 2-Mark]

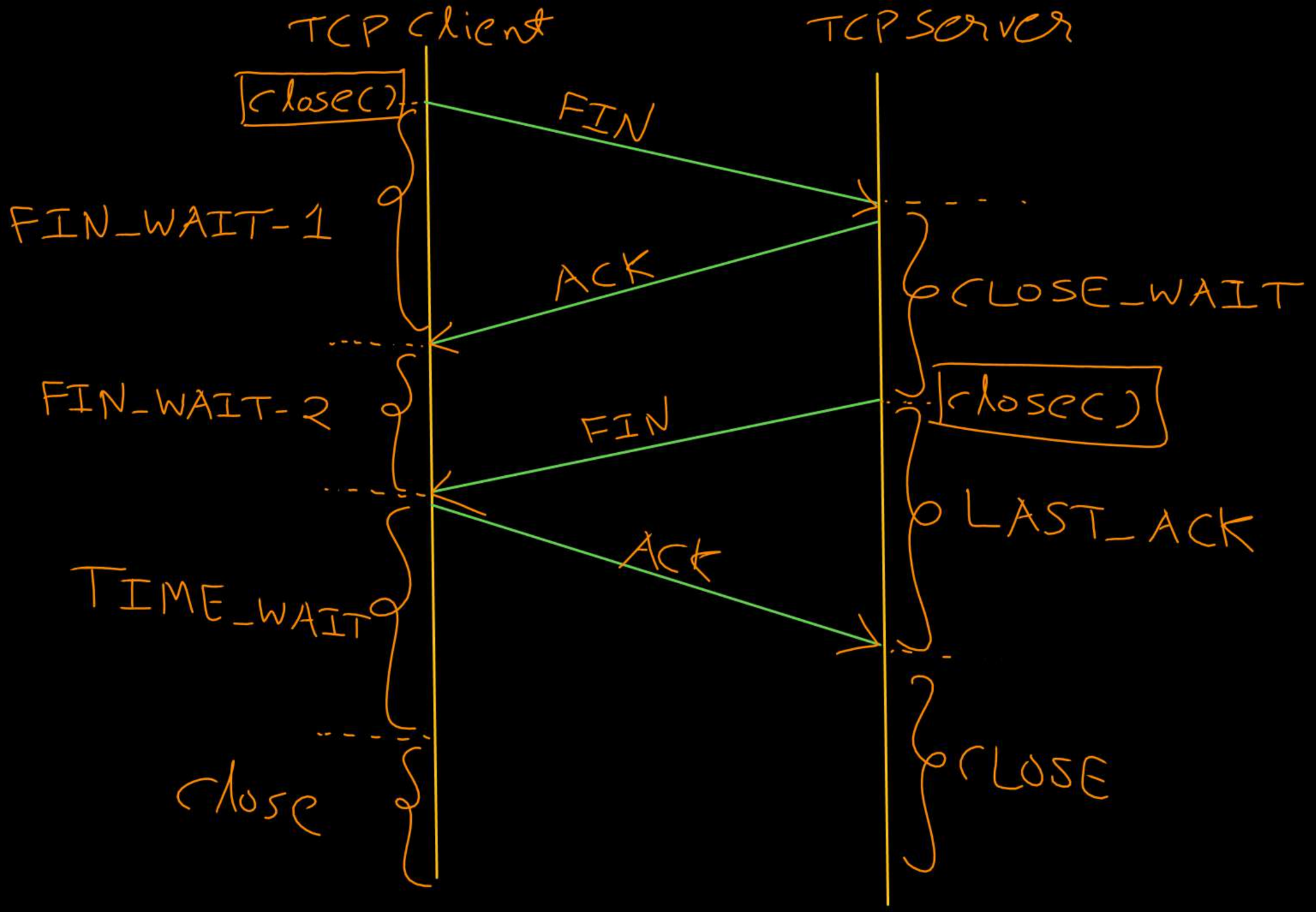
IIT-R

(A) LAST-ACK

(B) TIME-WAIT

(C) FIN-WAIT-1

(D) FIN-WAIT-2


TCP client
TCP server
close()
FIN
FIN_WAIT-1
ACK
CLOSE_WAIT
FIN_WAIT-2
close()
FIN
LAST_ACK
ACK
TIME_WAIT
close
CLOSE

Topic : Sockets

→ Socket is data structure inside the program

→ Both client and server programs communicate via a pair of sockets

Socket API :

→ Set of library function calls

→ Between Application Layer and Transport Layer

→ Door between application process and end-to-end transport protocol

Socket = End point of communication

Topic : Socket Programming / Network Programming

→ Client / server application that communicate using sockets

Process : Controlled by application developer

OSI Model : Mostly controlled by operating system

Topic : Types of Sockets

Stream Socket : SOCK_STREAM

→ Uses TCP protocol

→ Connection-oriented service

→ Reliable and byte stream oriented

Datagram Socket : SOCK_DGRAM

→ Uses UDP protocol

→ Connection-less service

→ Unreliable datagram

Topic : Creating a Socket

```
#include <sys/types.h>
#include <sys/socket.h>

int socket ( int domain, int type, int protocol );
```

domain : Protocol Family
[PF_INET, AF_INET, PF_UNIX etc]

type : Communication Protocol semantics
[SOCK_STREAM, SOCK_DGRAM]

protocol : Specify particular protocol, set "zero" as default

return : On success, a file descriptor for the new socket
On error, -1 is returned

```
sockfd = socket(PF_UNIX,
                SOCK_STREAM,
                0);
```

Sockfd

Socket
(Stream Socket)

Topic : Bind a name to a Socket

```
#include <sys/types.h>
#include <sys/socket.h>

int bind ( int sockfd, const struct sockaddr *addr, socklen_t addrlen );
```

(Server own IP & Port No.)

sockfd : file descriptor of the socket (Server Socket)

return : On success, zero
On error, -1 is returned

Topic : Listen for connections

```
#include <sys/socket.h>

int listen ( int sockfd, int backlog );
```

sockfd : file descriptor of the socket (server socket)

backlog : maximum length for queue of pending connections

return : On success, zero
On error, -1 is returned

Topic : Socket Programming with TCP

Server Process

TCP Server

1. socket()
2. bind()
3. listen()

→ Ready to listen client request

→ Welcome Socket

Topic : Initiate a connection

(Server IP & Port No.)

```
#include <sys/socket.h>

int connect ( int sockfd, const struct sockaddr *addr, socklen_t addrlen );
```

sockfd : file descriptor of the socket (client socket)

return : If connection succeeds, zero
On error, -1 is returned

Topic : Socket Programming with TCP

TCP Client		TCP Server
socket()		socket()
connect()	SYN (Connection Request) →	bind()
		listen()

Topic : Accept a connection

→ Creates a new connected (connection) socket

(client IP & Port)

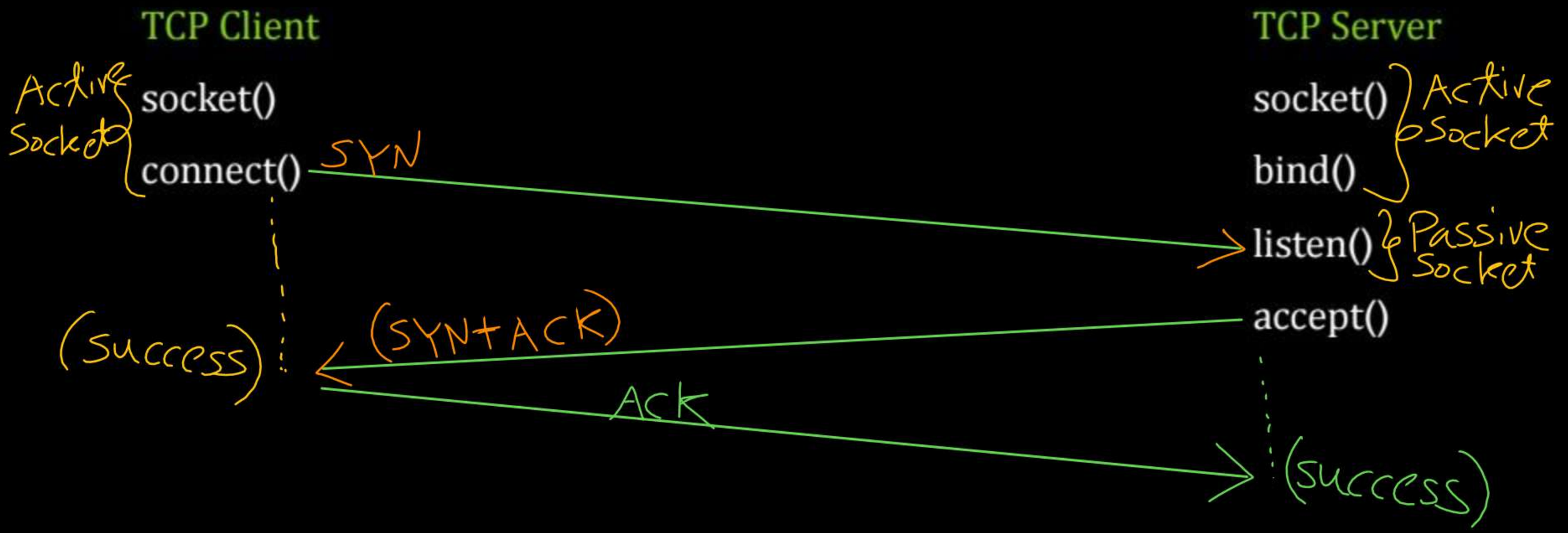
```
#include <sys/socket.h>

int accept ( int sockfd, struct sockaddr *_Nullable restrict addr,
                                socklen_t *_Nullable restrict addrlen );
```

sockfd : file descriptor of the socket (server socket)

return : On success, file descriptor for the accepted socket (connection Socket)
On error, -1 is returned

Topic : Socket Programming with TCP



Connection Oriented Demultiplexing:-

At server:-

① Welcome Socket : Identify
① Server IP + Port No.

② Connection Socket : Identify
① Server IP + Port No.
② Client IP + Port No.

Topic : Write to a socket

```
#include <unistd.h>

ssize_t write ( int sockfd, const void buf[.count], size_t count);
```

sockfd : file descriptor of the socket

return : On success, the number of bytes written is returned
On error, -1 is returned

Topic : Read from a socket

```
#include <unistd.h>

ssize_t read ( int sockfd, void buf[.count], size_t count);
```

sockfd : file descriptor of the socket

return : On success, the number of bytes read is returned
On error, -1 is returned

Topic : Socket Programming with TCP

TCP Client		TCP Server
socket()		socket()
connect()	SYN →	bind()
		listen()
	← SYN+ACK	accept()
	ACK →	
write()	Data →	read()
read()	←	write()

Topic : Close a socket

```
#include <unistd.h>

ssize_t close ( int sockfd );
```

sockfd : file descriptor of the socket (connection Socket)

return : On success, returns zero
On error, -1 is returned

Topic : Socket Programming with TCP

TCP Client	TCP Server
socket()	socket()
connect()	bind()
	listen()
	accept()
write()	read()
read()	write()
close() ← 4-way hand shaking →	close()

#Q. Which of the following system calls results in the sending of SYN packets?

[GATE-2008]

IISc

(A) socket

(B) bind

(C) listen

(D) connect

Ans: D

#Q. Which one of the following socket API functions converts an unconnected active TCP socket into a passive socket?

[GATE-2014, Set-2, 1-Mark]

IIT-KGP

(A) connect

(B) bind

(C) listen

(D) connect

Ans: C

#Q. Identify the correct order in which a server process must invoke the function calls accept, bind, listen, and recv according to UNIX socket API.

[GATE-2015, Set-2, 1-Mark]

(A) listen, accept, bind, recv

(B) bind, listen, accept, recv

(C) bind, accept, listen, recv

(D) accept, listen, bind, recv

Ans: B

IIT-K

TCP Server

1. socket
2. bind
3. listen
4. accept
5. recv

#Q. A client process P needs to make a TCP connection to a server process S. Consider the following situation: the server process S executes a socket(), a bind() and a listen() system call in that order, following which it is preempted. Subsequently, the client process P executes a socket() system call followed by connect() system call to connect to the server process S. The server process has not executed any accept() system call. Which one of the following events could take place?

[GATE-2008]

IISc

H.W.

(A) connect() system call returns successfully

(B) connect() system call blocks

(C) connect() system call returns an error

(D) connect() system call results in a core dump

2 mins Summary

Topic — Socket Programming

Topic — TCP Socket

Topic — ~~UDP Socket~~

THANK - YOU